



MALAPPURAM SAHODAYA APTITUDE TEST EXAMINATION 2024-25

School Level

Class: 10

Date:

Marks: 100

Name of the Student:

General instructions

- I. Read all the questions carefully.
- II. Attempt all the questions.
- III. QP 01 TO 40 (1MARK), QP41 TO 70 (2MARKS)

1 Read the passage given below and answer the following questions:

The concept of artificial intelligence has evolved rapidly, transitioning from a distant futuristic idea to a central aspect of modern technology. AI, particularly in the form of machine learning, has permeated various fields, including healthcare, finance, and transportation. One of AI's most striking advancements is its ability to mimic human decision-making processes. In many ways, AI can outperform humans in speed, accuracy, and efficiency. Yet, this remarkable development brings forth ethical and philosophical challenges. The primary concern revolves around the autonomy of AI systems and their capacity to make decisions without human oversight. While AI can handle complex tasks like diagnosing diseases or analyzing massive datasets, there is growing fear that its widespread adoption could lead to job displacement, privacy erosion, and even the loss of human agency. As the line between human intelligence and artificial intelligence blurs, society must grapple with the implications of creating systems that are, in some respects, more intelligent than their creators.

1. Which of the following best describes the current role of AI in various fields, as mentioned in the passage?
 - A) AI is primarily used in entertainment.
 - B) AI remains in the experimental phase.
 - C) AI is deeply integrated into healthcare, finance, and transportation.
 - D) AI is mostly used for educational purposes.
2. Why does the author highlight AI's ability to "mimic human decision-making processes"?
 - A) To show AI's limitations in understanding emotions
 - B) To emphasize its superiority over human intelligence
 - C) To demonstrate AI's capability to handle complex tasks
 - D) To argue for the reduction of AI usage in technology
3. Which of the following is implied as a potential downside of AI's widespread adoption?
 - A) Greater human freedom and agency
 - B) Improvement in job opportunities
 - C) Privacy erosion and job displacement
 - D) Reduction in technological dependence
4. The phrase "the line between human intelligence and artificial intelligence blurs" implies that:
 - A) AI and human intelligence are increasingly indistinguishable.
 - B) AI is quickly surpassing human intelligence in all areas.
 - C) Human intelligence is diminishing as AI advances.
 - D) AI will never match human intelligence.

II. Choose the correct option:

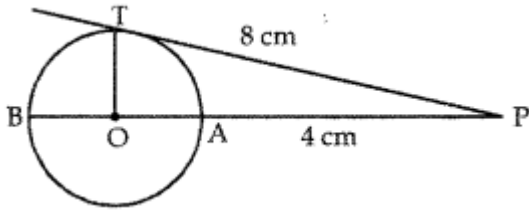
5. His speech was so _____ that it moved everyone in the audience to tears.
A) Mundane B) Pedantic C) Eloquent D) Irrelevant
6. The CEO kept moving the goalposts by changing the project requirements every week. What does the idiom "moving the goalposts" mean in this context?
A) Trying to improve the team's performance
B) Setting unrealistic expectations
C) Changing the rules or expectations unfairly
D) Encouraging the team to work harder
7. Which word is a synonym of "equine"?
A. Horse-like B. Fast C. Majestic D. Grazing
8. Choose the correct word to complete the sentence: "Neither the manager nor the employees _____ willing to accept the new policy."
A. was B. were C. had D. Is
9. Which option BEST completes the given sentence? "Her laughter filled the room, as contagious as _____."
A. a quiet thought C. a distant echo
B. a joyful melody D. a heavy burden
10. Fill in the blank: "If they had known about the meeting, they _____ attended."
A. would B. will C. would have D. Have
11. Two pea plants, one with round green seeds (RRyy) and another with wrinkled yellow (rrYY) seeds produce F1 progeny that have round, yellow (RrYy) seeds. When F1 plants are selfed, the F2 progeny will have a new combination of characters. Choose the new combination from the following
(i) Round, yellow (iii) Wrinkled, yellow
(ii) Round, green (iv) Wrinkled, green
A) (i) and (ii)
B) (i) and (iv)
C) (ii) and (iii)
D) (i) and (iii)
12. Contraceptive oral pills help in birth control process by
A Preventing ovulation
B Killing the ova
C Killing the sperms
D Forming barriers between sperm and ova
13. Most of the carbon dioxide in the blood is carried in the form of
A) Carbonic acid.
B) Bicarbonates.
C) Carbaminohaemoglobin.
D) Dissolved CO₂.
14. Read the following statements carefully:
(1) Energy transfer in the biotic world always proceeds from the autotrophs.
(II) Energy flow is unidirectional.
(III) Energy availability is maximum at the tertiary level.
(IV) There is a loss of energy from one trophic level to the other.
- Select the relevant statements for a forest ecosystem
A) I, II, III
B) I, II, IV
C) I, III, IV
D) II, III, IV

15. The correct arrangement of metals Al, Mg, Na and Zn in the decreasing order of their reactivity towards oxygen is
- A) Na > Al > Mg > Zn
B) Na > Mg > Al > Zn
C) Na > Al > Zn > Mg
D) Mg > Na > Al > Zn
16. A sample of soil is mixed with water and allowed to settle. The clear supernatant solution turns the pH paper yellowish orange. Which of the following would change the colour of this pH paper to greenish blue:
- A) lemon juice
B) vinegar
C) common salt
D) an antacid
17. The magnetic field strength of a solenoid can be increased by inserting which of the following materials as the core?
- A) copper
B) silver
C) iron
D) aluminium
18. The element 'X' makes an oxide with the formula $x_2 o_3$. This element will be in the same group as
- A) Na
B) Mg
C) Cl
D) Al
19. The functional group present in HCHO
- A) aldehyde
B) ketone
C) esters
D) hydroxyl
20. The device used for voltage regulator is
- A) schottky diode
B) zener diode
C) varactor diode
D) tunnel diode
21. Which pact resolved the issue of separate electorates for dalits between Gandhi and Ambedkar in 1932?
- (A) Lucknow pact (B) Nagpur pact (C) Poona pact (D) Surat pact
22. Which of the following is an example of holding together federations?
- A) India (B) USA (C) Switzerland (D) None of the above.
23. _____ used to say that religion can never be separated from politics. He believed that politics must be guided by ethics drawn from religion.
- A) C Rajagopalachari (B) Sardar Patel (C) Gandhiji (D) Nehru.
24. There are two statements marked as Assertion (A) and Reason (R). Mark your answer as per the codes provided below: Assertion (A): The share of agriculture in the Gross Domestic Product (GDP) is declining in India. Reason (R): Agriculture sector is facing problems of unirrigated land, increasing debts, seasonal employment etc.
- A) Both A and R are true but R is not the correct explanation of A.
B) Both A and R are true and R is the correct explanation of A.
C) A is false but R is true.
D) A is true but R is false.
25. In which country is the Grameen Bank meeting the credit needs of over 6 million poor people?
- A) Bhutan (B) Sri Lanka (C) Bangladesh (D) Nepal
26. Which of the following option can be a restriction for foreign trade ?
- A) Sale tax (B) Import taxes (C) Local trade taxes (D) Quality control
27. Which of the following statements is not correct in the context of the Non - Cooperation Movement?
- A) Gandhiji was the main force behind the non - cooperation movement
B) The council elections were boycotted even by the Justice Party of Madras
C) The movement aimed to involve people from all sections of society.
D) Thousands of students left government - controlled schools; headmasters and teachers resigned; lawyers gave up their practice.
28. The Directive Principles of State Policy are:
- A) . Enforceable by courts

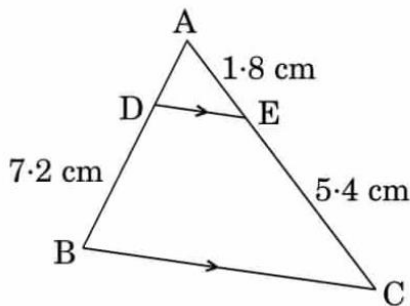
- B) . Non-enforceable by courts
 C) . Directly applicable to citizens
 D) . Binding on the central government only
29. The "3Rs" principle in waste management stands for:
 A) . Reduce, Reuse, Recycle
 B) . Recycle, Reuse, Reduce
 C) . Reuse, Reduce, Recycle
 D) . Recycle, Reduce, Reuse
30. Which of the following countries was not a founding member of the United Nations Organization (UNO)?
 A) . France
 B) . Great Britain
 C) . Germany
 D) . United States
31. What is the term used to describe the symbiotic relationship between a fungus and a plant root?
 A. Parasitism
 B. Commensalism
 C. Mutualism
 D. Predation
32. What is the largest living organism on Earth?
 A. Blue Whale
 B. Giant Sequoia Tree
 C. Amazon Rainforest
 D. African Elephant
33. Analyze the impact of a 2023 Nobel Prize winner in Literature on the literary landscape. How has their work influenced themes, styles, or cultural perspectives?
 A. Olga Tokarczuk
 B. Alice Munro
 C. Kazuo Ishiguro
 D. Bob Dylan
34. Evaluate the economic and cultural impact of hosting the Summer Olympics on a host city or country. Use examples from recent Olympic Games to support your analysis. Which country hosted the 2024 Summer Olympics?
 A. Paris, France
 B. Los Angeles, USA
 C. Brisbane, Australia
 D. Tokyo, Japan
35. This ancient amphitheater is located in which city?
 A. Rome, Italy.
 B. Athens, Greece
 C. Cairo, Egypt
 D. Istanbul, Turkey
36. The internet is based on which technology?
 A. Radio Waves
 B. Television Signals
 C. Satellite Communication
 D. Packet Switching
37. What is the term used to describe the process of converting sunlight into electricity?
 A. Photosynthesis
 B. Solar Energy
 C. Photovoltaic Effect
 D. Solar Thermal Energy
38. The economic and diplomatic significance of India hosting a major global event in 2024. This event shape India's international standing and influence global discourse on key issues. Which global event was held in India in 2024?
 A. G20 Summit
 B. COP28 Climate Conference
 C. BRICS Summit
 D. ASEAN Summit
39. Who won the 2024 US Presidential Election?
 A. Joe Biden
 B. Donald Trump
 C. Ron DeSantis
 D. Mike Pence
40. Time management skills involve:
 A. Procrastinating
 B. Overcommitting oneself
 C. Setting priorities and managing time effectively
 D. Ignoring deadline



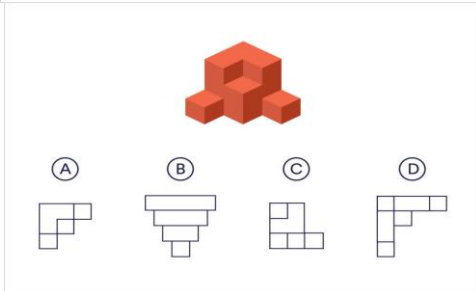
41. In the given figure, O is the centre of the circle, PT is the tangent and PAB is the secant passing through centre O. If PT = 8 cm and PA = 4 cm, then find the radius of the circle.



- A) 9cm B) 10cm C) 7cm D) 6cm
42. Find the value of k, for which one root of the quadratic equation $kx^2 - 14x + 8$ is 6 times the other.
A) 3 B) -3 C) 0 D) 2
43. The least number that is divisible by all the numbers from 1 to 10 (both inclusive)
A) 10 B) 100 C) 504 D) 2520
44. The zeros of the polynomial $x^2 - 3x - m(m+3)$ are
A) m, m+3 B) -m, m+3 C) m, -(m+3) D) -m, -(m+3)
45. The point on the X axis which is equidistant from the points A (-2, 3) and B (5, 4) is
A) (0, 2) B) (2, 0) C) (3, 0) D) (-2, 0)
46. A solid sphere is cut into two hemispheres. The ratio of the surface areas of sphere to that of two hemispheres taken together is
A) 1:1 B) 1:4 C) 2:3 D) 3:2
47. Two dice are thrown together. The probability that sum of the two numbers will be a multiple of 4 is
A) $1/2$ B) $1/3$ C) $1/8$ D) $1/4$
48. If the 3rd and the 9th terms of an A P are 4 and -8 respectively, which term of this A P is zero?
A) 5 B) 4 C) 6 D) 8
49. In the figure, $DE \parallel BC$. Find the length of side AD, given that AE = 1.8 cm, BD = 7.2 cm and CE = 5.4 cm.



- A) 2.4 cm B) 3.7cm C) 4.2cm D) 7.3cm
50. If $\sin A = 1/3$, then find the value of $9 \cot^2 A + 9$
A) 8. B) 9 C) 81 D) 18
51. What would the 3D shape look like from above?



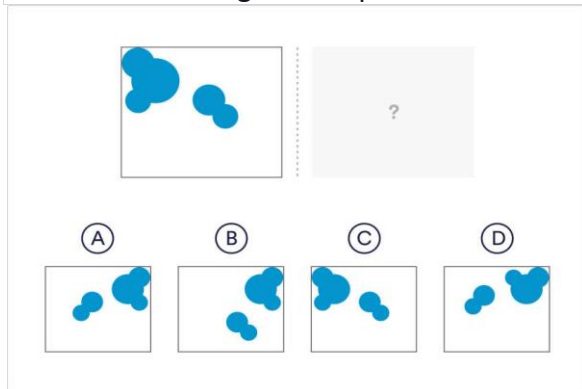
A)A

B)B

C)C

D)D

52. Which of the given shapes is the correct mirror image?



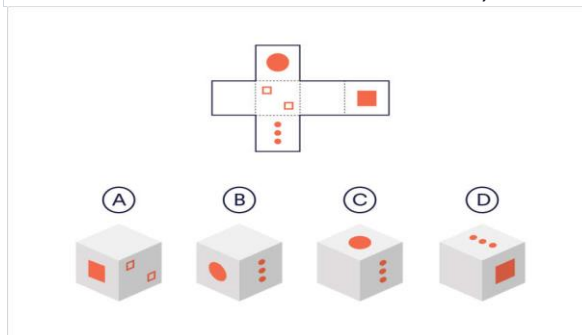
A)A

B)B

C)C

D)D

53. If the net was folded into a cube, which of the given shapes would it look like?



A)A

B)B

C)C

D)D

54. Oak: Tree :: Rose: - ?

- A) Pen B) Car C) Flower D) Food

55. Life is related to Death in the same way as Hope is related to?

- A) Sad B) Despair C) Pain D) Cry

56. Directions: The following question consists of two words that have a certain relationship to each other, followed by four lettered pairs of words. Select that lettered pair which has the same relationship as the original pair of words.

Calligraphy: Writing

- A) Music: Song B) Lyric: Poem C) Drama: Prose D) Chapter: Stanza

57. Find the odd man out.

1, 4, 9, 16, 23, 25, 36

A) 4

B) 25

C) 23

D) 36

58. There are two examinations rooms A and B. If 10 students are sent from A to B, then the number of students in each room is the same. If 20 candidates are sent from B to A, then the number of students in A is double the number of students in B. The number of students in room A is:

A) 20

B) 80

C) 100

D) 200

59. In a 100m race, A beats B by 20 m and B beats C by 20 m. What is the distance by which A beats C?

A) 64 m

B) 24 m

C) 25 m

D) 36 m

60. A man has 100 coins consisting of ₹1 and ₹2 coins. The total value of these coins is ₹160. How many ₹2 coins does he have?

A) 60

B) 30

C) 40

D) 50

61. In how many different ways can the letters of the word 'DETAIL' be arranged in such a way that the vowels occupy only the odd positions?

A) 32

B) 48

C) 36

D) 60

62. The price of commodity X increases by 40 paise every year, while the price of commodity Y increases by 15 paise every year. If in 2001, the price of commodity X was Rs. 4.20 and that of Y was Rs. 6.30, in which year commodity X will cost 40 paise more than the commodity Y?

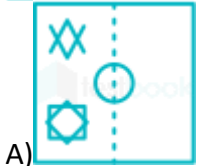
A) 2010

B) 2011

C) 2012

D) 2013

63. If the transparent paper given below is folded at the dotted line what would be the result of the folding:



B)



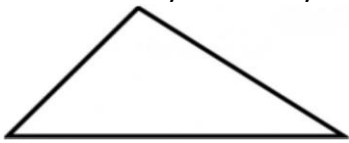
C)



D)



64. How many lines of symmetry does the following figure (Scalene triangle) have?



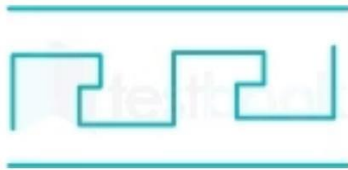
A) 3

B) 2

C) 1

D) 0

65. The order of rotational symmetry in the figure given below is.



A) 4

B) 2

C) 1

D) Infinitely many

66. In a circle every ___ is a line of symmetry

A) diameter

B) both diameter and chord

C) chord

D) None of those

67. The letter F has

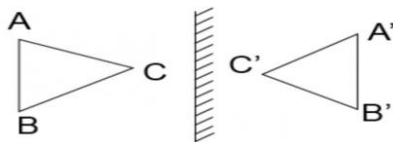
A) no line of symmetry

B) one vertical line of symmetry

C) one horizontal line of symmetry

D) two lines of symmetry

68. The two figures are mirror images of each other. Choose the correct option.



A) Both are equal

B) Both are similar

C) Both option (a) and option (b) are correct

D) None of the above

69. Arrange the words given below in a meaningful sequence.

1. Police 2. Punishment 3. Crime 4. Judge 5. Judgment

A) 3, 1, 2, 4, 5

B) 1, 2, 4, 3, 5

C) 5, 4, 3, 2, 1

D) 3, 1, 4, 5, 2

70. A, P, R, X, S, and Z are sitting in a row. S and Z are in the centre. A and P are at the ends. R is sitting to the left of A. Who is to the right of S?

A) R

B) X

C) Z

D) P
